INT421—Digital Signal Processing

The z-Transformation-I

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Topics of Last Lecture

- Definition of Fourier transformation.
- Definition of inverse Fourier transformation.
- Condition for a signal to have DTFT.

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1. The z-transform
2. Relation between z-Transform and Fourier Transform
3. Convergence Region
4. Properties of z-Transform
5. Frequency Response
6. Summary

The z-transform

- z-transformation is important for analyzing discrete signal/systems.
- For a given signal $x[n]$, its z-transformation is defined by:

$$Z(X[n]) = X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

  where z is a continuous complex variable.
- The z-transform defined above is called **one-sided** z-transform and is meaningful for causal signals.
- For non-causal signals, the summation should start from $-\infty$, and the z-transform so defined is called **two-sided** z-transform.

Quiz

Examples

Find the z-transform for $\delta[n]$, $u[n]$, and $\alpha^n u[n]$

Quiz

Solution

Find the z-transform for $\delta[n]$, $u[n]$, and $\alpha^n u[n]$.

1. $Z(\delta[n]) = \delta[0] + \delta[1]z^{-1} + \delta[1]z^{-1} + \cdots + \delta[n]z^{-n} + \cdots = 1, \quad$ for any z

Quiz

Solution

Find the z-transform for $\delta[n]$, $u[n]$, and $\alpha^n u[n]$.

1. $Z(\delta[n]) = \delta[0] + \delta[1]z^{-1} + \delta[1]z^{-1} + \cdots + \delta[n]z^{-n} + \cdots = 1, \quad$ for any z
2. $Z(u[n]) = u[0] + u[1]z^{-1} + u[2]z^{-2} + \cdots + u[n]z^{-n} + \ldots$
   $= 1 + z^{-1} + z^{-2} + z^{-3} + \cdots + z^{-n} = \frac{1}{1-z^{-1}} == \frac{z}{z-1}, \quad$ for all $|z| > 1$

Quiz

Solution

Find the z-transform for $\delta[n]$, $u[n]$, and $\alpha^n u[n]$.

1. $Z(\delta[n]) = \delta[0] + \delta[1]z^{-1} + \delta[1]z^{-1} + \cdots + \delta[n]z^{-n} + \cdots = 1, \quad$ for any z
2. $Z(u[n]) = u[0] + u[1]z^{-1} + u[2]z^{-2} + \cdots + u[n]z^{-n} + \ldots$
   $= 1 + z^{-1} + z^{-2} + z^{-3} + \cdots + z^{-n} = \frac{1}{1-z^{-1}} == \frac{z}{z-1}, \quad$ for all $|z| > 1$
3. $Z(\alpha^n u[n]) = u[0] + \alpha u[1]z^{-1} + \alpha^2 u[2]z^{-2} + \cdots + \alpha^n u[n]z^{-n} + \ldots$
   $= 1 + \alpha z^{-1} + \alpha^2 z^{-2} + \alpha^3 z^{-3} + \cdots + \alpha^n z^{-n} = \frac{1}{1-\alpha z^{-1}} = \frac{z}{z-\alpha}, \quad$ for all $|z| > |\alpha|$

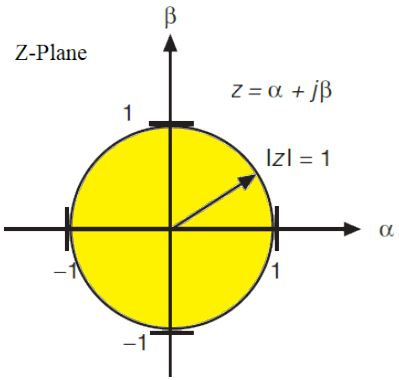
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Relation between z-Transform and Fourier Transform

- Fourier transform is the z-transform on the **unit circle**.
- The values of the z-transform on the unit circle equals to those of the Fourier transform of the same signal.

$$X(e^{j\omega}) = X(z)|_{e^{-j\omega} or |z|=1}$$



Relation between z-Transform and Fourier Transform

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- We may obtain the Fourier transform from the z-transform by making the substitution $z = e^{j\omega}$. This corresponds to restricting $|z| = 1$. Also, with $z = re^{j\omega}$.

$$X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n](re^{j\omega})^{-n} = \sum_{n=-\infty}^{\infty} (x[n]r^{-n})e^{j\omega n}$$

- That is, the z-transform is the Fourier transform of the sequence $(x[n]r^{-n})$
- For $r = 1$ this becomes the Fourier transform of $x[n]$. So, the Fourier transform therefore corresponds to the z-transform evaluated on the unit circle:

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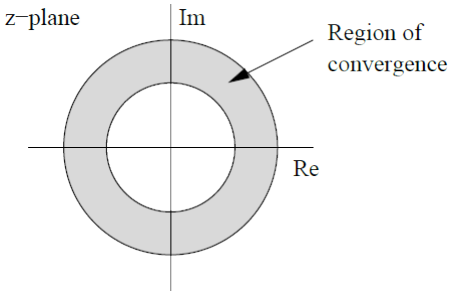
Convergence Region

- Fourier transform exists (converges) if and only if the signal is absolutely summable.
- Similarly, the z-transform does not converge for all sequences or for all values of z.
- The region in which the z-transform exists is called the **convergence region**.
- The set of values of z for which the z-transform converges is called the **region of convergence (ROC)**.
- z-transformation is a more powerful tool for analyzing signals.
- The z-transform therefore exists (or converges) if

$$X(z) = \sum_{n=-\infty}^{\infty} (x[n]z^{-n}) < \infty$$

Convergence Region

- For the existence of the z-transform. The ROC therefore consists of a ring in the z-plane:



- If the ROC includes the unit circle $|z| = 1$, then the Fourier transform will converge.

Quiz

Examples

Find the z-transform for the signal $\delta[n-1]$ and its convergence condition

Quiz

Solution

Find the z-transform for the signal $\delta[n-1]$ and its convergence condition

1. $Z(\delta[n-1]) = \delta[-1] + \delta[0]z^{-1} + \delta[1]z^{-2} + \cdots + \delta[n-1]z^{-n} + \ldots$

$$Z(\delta[n-1]) = \sum_{n=0}^{\infty} \delta[n-1]z^{-n} = \delta[0]z^{-1} = z^{-1}$$

2. **ROC**: *entire z-plane except $z = 0$.*

Quiz

Examples

Find the z-transform for the signal $x[n] = u[n-1]$ and its convergence condition

Quiz

Solution

Find the z-transform for the signal $x[n] = u[n-1]$ and its convergence condition

1. *The signal $x[n] = u[n-1]$ is:*

$$x[n] = \delta[1] + \delta[2] + \delta[3] + \delta[4] + \dots$$
$$Z(x[n]) = z^{-1} + z^{-2} + z^{-3} + z^{-4} + \dots$$
$$Z(x[n]) = \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \cdots = \frac{1}{z}(1 + \frac{1}{z} + \frac{1}{z^2} + \dots) = \frac{1}{z}\left[\frac{1}{1 - z^{-1}}\right] = \frac{1}{z-1}$$

2. **ROC**: $|z| > 1$

This is a geometric series: $$s = \sum_{n=0}^{n-1} \alpha r^n = \alpha \frac{1 - r^n}{1 - r} = \lim_{n \to \infty} \frac{\alpha}{1 - r}$$

Quiz

Examples

Find the z-transform for the signal $x[n] = [\underset{\uparrow}{1}, 2, 5, 7, 0, 1]$ and its convergence condition

Quiz

Solution

Find the z-transform for the signal $x[n] = [\underset{\uparrow}{1}, 2, 5, 7, 0, 1]$ *and its convergence condition*

1. *The signal* $x[n] = [\underset{\uparrow}{1}, 2, 5, 7, 0, 1]$ *is:*

$$x[n] = \delta[n] + 2\delta[n-1] + 5\delta[n-2] + 7\delta[n-3] + \delta[n-5]$$
$$Z(x[n]) = 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$$
$$Z(x[n]) = 1 + \frac{2}{z} + \frac{5}{z^2} + \frac{7}{z^3} + \frac{1}{z^5}$$

2. **ROC**: *entire z plane except* $z = 0$ *and* $z = \infty$

Quiz

Examples

Find the z-transform for the signal $\alpha^n u[n]$ and its convergence condition

Quiz

Solution

Find the z-transform for the signal $\alpha^n u[n]$ and its convergence condition

1. $Z(\alpha^n u[n]) = u[0] + \alpha u[1] z^{-1} + \alpha^2 u[2] z^{-2} + \cdots + \alpha^n u[n] z^{-n} + \ldots$

$$Z(\alpha^n u[n]) = \sum_{n=-\infty}^{\infty} \alpha^n u[n] z^{-n} = \sum_{n=-\infty}^{\infty} (\alpha z^{-1})^{-n} = \frac{1}{1-\alpha z^{-1}} = \frac{z}{z-\alpha}$$

2. *Convergence requires that*

$$\sum_{n=-\infty}^{\infty} |\alpha z^{-1}|^{-n} < \infty$$

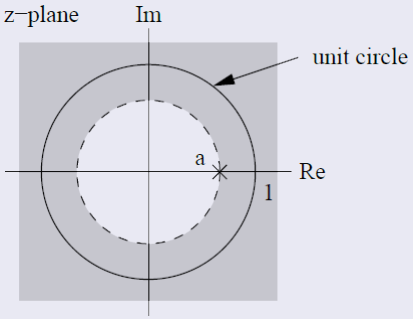
3. *Which is only the case if $|\alpha z^{-1}| < 1$, or equivalently* **ROC**: $|z| > |\alpha|$.

Quiz

Solution

Find the z-transform for the signal $\alpha^n u[n]$ and its convergence condition

1. *The z-transform has a region of convergence for any finite value of α.*



Quiz

Examples

Find the z-transform for the a step function

$$x[n] = \begin{cases} 0.5, & 0 \leq n < 5 \\ 0, & otherwise \end{cases}$$

Quiz

Solution

Find the z-transform for the a step function

$$x[n] = \begin{cases} 0.5, & 0 \leq n < 5 \\ 0, & otherwise \end{cases}$$

$$Z(x[n]) = \sum_{n=0}^{\infty} x[n] z^{-n} = 0.5 \sum_{n=0}^{4} z^{-n}$$

$$Z(x[n]) = 0.5(1 + z^{-1} + z^{-2} + z^{-3} + z^{-4})$$

$$Z(x[n]) = 0.5 \frac{1 - r^{-5}}{1 - z^{-1}}$$

Remarks

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$\|z\| > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$\|z\| < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\|z\| > \|a\|$
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$\|z\| < \|a\|$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| > \|a\|$
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| < \|a\|$
$\begin{cases} a^n & 0 \le n \le N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$\|z\| > 0$
$\cos(\omega_0 n)u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$\|z\| > 1$
$r^n \cos(\omega_0 n)u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2 z^{-2}}$	$\|z\| > r$

- It is important to specify the **convergence region** in the results.
- If we use the left table, we can find the z-transform of more complicated signals.

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Linearity

if $Z(x_1[n]) = X_1(z)$ and $Z(x_2[n]) = X_2(z)$ then

$$Z(ax_1[n] + bx_2[n]) = aX_1(z) + bX_2(z)$$

Example

Find the z-transform of the sequence defined by $x[n] = u[n] - (0.5)^n u[n]$

Solution

Applying the linearity of the z-transform, we have

$$\begin{aligned} Z(x[n]) = Z(x[n]) &= Z(u[n] - (0.5)^n u[n]) \\ &= Z(u[n]) - Z((0.5)^n u[n]) \\ &= \frac{z}{z-1} - \frac{z}{z-0.5} \end{aligned}$$

Linearity

Example

Find the z-transform of the sequence defined by $x[n] = \cos[\omega n + \phi]u[n]$

Solution

Applying Euler's identity

$$x[n] = \frac{1}{2}\left(e^{j(\omega n+\phi)} + e^{-j(\omega n+\phi)}\right)u[n] = \frac{1}{2}e^{j\phi}(e^{j\omega})^n u[n] + \frac{1}{2}e^{j\phi}(e^{-j\omega})^n u[n]$$

Applying previous example with $\alpha = e^{\pm j\omega}$ and linearity:

$$X(z) = \frac{\frac{1}{2}e^{j\phi}}{1 - e^{j\omega}z^{-1}} + \frac{\frac{1}{2}e^{-j\phi}}{1 - e^{-j\omega}z^{-1}} = \frac{\cos\phi - z^{-1}(\cos(\omega - \phi))}{1 - 2z^{-1}\cos(\omega) + z^{-2}}$$

Time Shifting/Translation

if $x[n] = 0$ for $n < 0$ and $Z(x[n]) = X(z)$ then $Z(x[n+m]) = z^m \left[X(z) - \sum_{n=0}^{m-1} x[n] z^{-n} \right]$ and $Z(x[n-m]) = z^{-m} X(z)$

Example

Find the z-transform of the sequence defined by $x[n] = (0.5)^{(n-5)} u[n-5]$

Solution

$$Z((0.5)^5 u[5]) = \frac{1}{1 - 0.5z^{-1}}$$

Applying the time shifting property of the z-transform, we have

$$Z(x[n]) = z^{-5} Z((0.5)^5 u[5]) = \frac{z^{-5}}{1 - 0.5z^{-1}} = \frac{z^{-4}}{z - 0.5}$$

Differentiation of X(z)

if $X(z)$ is the z-transformation of $x[n]$, then $Z(nx[n]) = -z\frac{dX(z)}{dz}$

Example

Find the z-transform of the sequence defined by $r[n] = nu[n]$

Solution

We Know that:

$$Z(u[n]) = \frac{1}{1 - z^{-1}}$$

Applying the differentiation property of the z-transform, we have

$$Z(r[n]) = Z(nx[n]) = -z\frac{dX(z)}{dz} = \frac{z^{-1}}{(1 - z^{-1})^2}$$

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Convolution

Convolution in time domain is equal to the multiplication in frequency domain and vice versa. if $x[n] \leftrightarrow X(z)$ and $y[n] \leftrightarrow Y(z)$ then

$$x[n] * y[n] \leftrightarrow X(z)Y(z)$$

Proof:

$$Z(x[n] * h[n]) = \sum_{n=-\infty}^{\infty} \left\{ \sum_{k=-\infty}^{\infty} x[k]h[n-k] \right\} z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x[k]z^{-n} \left\{ \sum_{k=-\infty}^{\infty} h[n-k]z^{-n} \right\} = X(Z)H(Z)$$

Convolution

Example

Consider the two sequences $x_1[n] = 3\delta[n] + 2\delta[n-1]$ and $x_2[n] = 2\delta[n] - \delta[n-1]$.

- Find the z-transform of convolution

$$x[n] = x_1[n] * x_2[n]$$

- Determine the convolution sum using the z-transform

Solution

- $Z(x_1[n]) = 3 + 2z^{-1}$ *and* $Z(x_2[n]) = 2 - z^{-1}$, *applying the multiplication of the* z*-transform, we have*

$$X(z) = Z(x_1[n] * x_2[n]) = 6 + z^{-1} - 2z^{-2}$$

- $x[n] = Z^{-1}(X(z)) = 6\delta[n] + \delta[n-1] - 2\delta[n-2]$

Convolution

Example

- Find the z-transform of convolution $x[n] = u[n] * u[n-1]$

Solution

- *The z-transform for* $u[n]$ *and* $u[n-1]$ *are*

$$Z(u[n]) = \frac{1}{1-z^{-1}} \quad \textit{and} \quad Z(u[n-1]) = z^{-1}\frac{1}{1-z^{-1}}$$

then

$$Z(x[n]) = \frac{z^{-1}}{(1-z^{-1})^2}$$

- *using the time-shift property. So* $x[n] = nu[n]$ *from previous example of differentiation.*
- *ROC for both* $u[n]$ *and* $u[n-1]$ *is* $|z| > 1$. *Same ROC for their convolution.*

Quiz

Problem

Construct the block diagram for the discrete-time systems whose input-output relations are described by the following difference equations

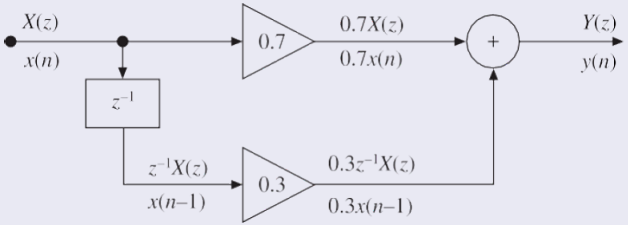
1. $y[n] = 0.7x[n] + 0.3x[n-1]$,
2. $y[n] = 0.5y[n-1] + 0.8x[n] + 0.4x[n-1]$.

Quiz

Solution

1. *Given* $y[n] = 0.7x[n] + 0.3x[n-1]$, *the system may be realized by using the difference equation directly or by using the z-transformed version of that. Taking Z-transform on both sides of the given difference equation, we have*

$$Y(z) = 0.7X(z) + 0.3z^{-1}X(z)$$



Quiz

Solution

1. *Given* $y[n] = 0.5y[n-1] + 0.8x[n] + 0.4x[n-1]$ *is represented by z-transform, we have*

$$Y(z) = 0.5z^{-1}Y(z) + 0.8X(z) + 0.4z^{-1}X(z)$$

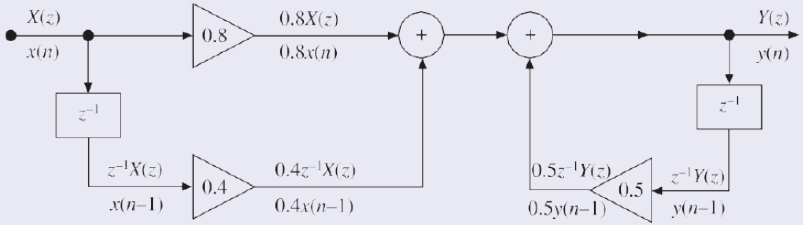


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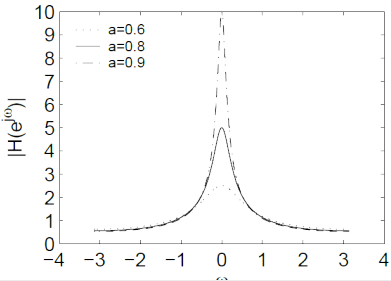
Frequency Response

The *frequency response* of the system can be obtained by evaluating the transfer function on the unit circle ($z = e^{j\omega}$). Actually can be obtained by replacing z with $e^{j\omega}$ in the transfer function. For example, frequency response for next transfer function

$$\frac{Y(z)}{X(z)} = \frac{1}{1 - az^{-1}}$$

$$H(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$

$$|H(e^{j\omega})| = \frac{1}{\sqrt{(1 - ae^{-j\omega})((1 - ae^{j\omega}}} = \frac{1}{\sqrt{1 - 2\cos\omega + a^2}}$$



Quiz

Examples

Given the digital system, with a sampling rate of $8,000$ Hz,

$$y[n] = 0.5x[n] + 0.5x[n-2], \quad \text{for all integer n.}$$

Determine and plot the magnitude and phase frequency responses. Determine the filter type.

Solution

Computing the transfer function of digital system,

$$H(z) = \frac{Y(z)}{X(z)} = 0.5 + 0.5z^{-2}$$

$$H(e^{j\omega}) = 0.5 + 0.5e^{-2j\omega} = 0.5(1 + \cos 2\omega) - j0.5\sin(2\omega)$$

$$|H(e^{j\omega})| = 0.5\sqrt{(1 + \cos 2\omega)^2 + (0.5\sin(2\omega))^2}, \angle H(e^{j\omega}) = \tan^{-1}\left(\frac{-\sin(2\omega)}{(1 + \cos 2\omega)}\right)$$

Ω (radians)	$f = \frac{\Omega}{2\pi} f_s$ (Hz)	$\lvert H(e^{j\Omega})\rvert$	$\lvert H(e^{j\Omega})\rvert_{dB}$	$\angle H(e^{j\Omega})$
0	0	1.000	0 dB	0.00^0
0.25π	1000	0.707	-3.0102 dB	-45.00^0
0.50π	2000	0.0	$-\infty$ dB	-90.00^0
0.75π	3000	0.707	-3.0102 dB	45.00^0
1.00π	4000	1.000	0 dB	0.00^0

Quiz

Solution

The filter type is bandstop filter.

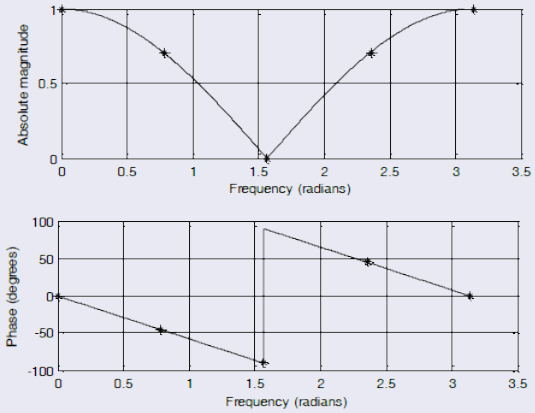


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Summary

- The z-transformation
- Convergence region of z-transform.
- Relation between z-transform and Fourier transform.
- Relation between z-transform and difference equation.
- Transfer function (system function).
- Frequency response.

Any Questions?

